John J. Podhiny

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Summary of Qualifications Over 8 years of experience of progressively increasing aerospace engineering responsibilities. Strong technical foundation with excellent communication and leadership skills.

Experience

Materials Research & Design, Inc. Wayne, PA

Materials Research & Design (MR&D) is a small consulting business founded in 1996, which provides research, analysis and design services to the advanced materials community, focusing primarily on composite materials for the aerospace industry. Primary customers are DoD Agencies, NASA, and aerospace prime contractors such as Boeing, Lockheed-Martin and Northrop-Grumman. MR&D employs ~20 full-time engineers, and has annual revenue of ~$6M (*www.m-r-d.com*).

Senior Research Engineer (December 2006 – Current)

- Manage the successful completion of projects for various aerospace customers by performing advanced technical research and analysis, generating written reports, and giving oral presentations to and personally interacting with clients.
- Assume full personal responsibility for obtaining quality results that meet strict deadlines.
- Work with minimal director supervision by making informed decisions on project direction.
- Apply sophisticated finite element modeling techniques to investigate and predict the behavior of complex thermal and structural aerospace vehicle systems, frequently developing methods that are completely new for MR&D, including the following:
  - Generated two Fortran subroutines to model the thermal and structural behavior of the unconventional polymer-based Space Shuttle Crack Repair Material.
  - Generated a Fortran subroutine to model the oxidation behavior of carbon-fiber-reinforced silicon-carbide matrix (C/SiC) composite materials.
  - Developed and applied the fracture-mechanics-based finite element analysis process that allows a comprehensive crack growth assessment for the Space Shuttle Tile Overlay Repair.
  - Define and assess the results of fully-automated parametric studies for numerous primary research problems.
- Regularly give oral and poster presentations at technical conferences.
- Generate proposals for various US Government agencies to develop innovative concepts related to mathematical modeling of composite materials.
- Provide technical and project-related consultation for other Senior Engineers.
- Mentor, advise and direct junior engineers.

Research Engineer (December 2000 – December 2006)

- Extensively use ABAQUS and ANSYS finite element analysis applications to perform random vibration analyses, structural analyses, thermal stress analyses, and heat transfer analyses, primarily on components made from orthotropic materials.
- Develop closed form analytical solutions to verify numerical analysis results.
- Use micromechanics tools to calculate composite thermo-elastic properties and strengths.
- Generate written reports and oral presentations to provide progress updates for clients.

Selected Consulting Projects:

- Design and analysis of refractory composite wing leading edges cooled by heat pipes (work performed for Air Force Research Laboratory).
- Design and analysis of C/SiC ceramic matrix composite Plug Repair and Rigid Overwrap Repair components for use on the Space Shuttle (work performed for ATK, Inc. and NASA).

- Analysis of a ceramic matrix composite bladetrack component for use in a high-performance turbine engine, taking into account thermal cycling and material degradation due to an oxidizing environment (work performed for Allison Advanced Development Company).

Computer Applications

Proficient in ABAQUS; ANSYS; Fortran; MathCAD; Matlab; MS Office (Excel, Outlook, Project, PowerPoint, Word); MS Windows XP.

Education

Villanova University Villanova, PA

Bachelor of Mechanical Engineering, Summa Cum Laude, May 2000 (3.94 GPA).

Master of Science in Mechanical Engineering, December 2004 (4.00 GPA).

Focused on Analysis of Composite Materials & Structures, Finite Element Analysis, Elasticity, Heat Transfer and Fluid Dynamics.

Doctor of Philosophy in Engineering, Current Candidate, (4.00 Current GPA).

Focusing on Energy Systems, Thermodynamics and Thermal-Fluids Science.

Certification

Passed Pennsylvania EIT / FE Examination in April 2000; score of 86.

Conference Presentations *Lead author and presenter for the following conference papers:*

"Development and Preliminary TGA Correlation Results of a Finite Element Based Durability Model for CMC Materials," 32nd Annual Conference on Composites, Materials & Structures; Daytona Beach, FL; January 2008.

"Formulation of Mathematical Model for Reentry Performance of Crack Repair Material for Space Shuttle Reinforced Carbon-Carbon," 31st Annual Conference on Composites, Materials & Structures; Daytona Beach, FL; January 2007.

"Environmental Life Modeling Development for C/SiC Composites," 31st Annual Conference on Composites, Materials & Structures; Daytona Beach, FL; January 2007.

"Math Model of Reentry Performance of Coating Crack Repair Material for Space Shuttle Reinforced Carbon-Carbon," JANNAF Joint Conference for Combustion, Airbreathing Propulsion, and Propulsion Systems Hazards; San Diego, CA; December 2006.

"Design and Analysis of Heat Pipe Cooled Refractory Composite Leading Edges," 2006 National Space and Missile Materials Symposium; Orlando, FL; June 2006.

"Design Considerations and Development Efforts for ATK C/SiC Intermediate Size Repair Concept," 29th Annual Conference on Composites, Materials & Structures; Cocoa Beach, FL; January 2005.

"Overview and Technical Accomplishments for C/SiC Large Size Repair Concept," 29th Annual Conference on Composites, Materials & Structures; Cocoa Beach, FL; January 2005.

"Design and Analysis of CMC Materials for AADC Turbine Engine Bladetrack," 28th Annual Conference on Composites, Materials & Structures; Cocoa Beach, FL; January 2004.

"Thermal-structural Analysis and Design of Innovative Thermal Protection Materials," 27th Annual Conference on Composites, Materials & Structures; Cocoa Beach, FL; January 2003.

Activities

- Part-time private tutor for high school students, specializing in calculus, algebra, physics, chemistry and SAT preparation; since September 2002.
- Hobbies: golf, softball, hockey, gas-powered radio controlled cars, numerous automotive related activities (reading, repairs, maintenance – oil changes, timing belt replacements, clutch and brake jobs, etc.).